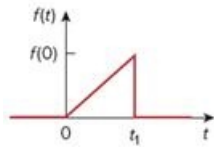


Problem

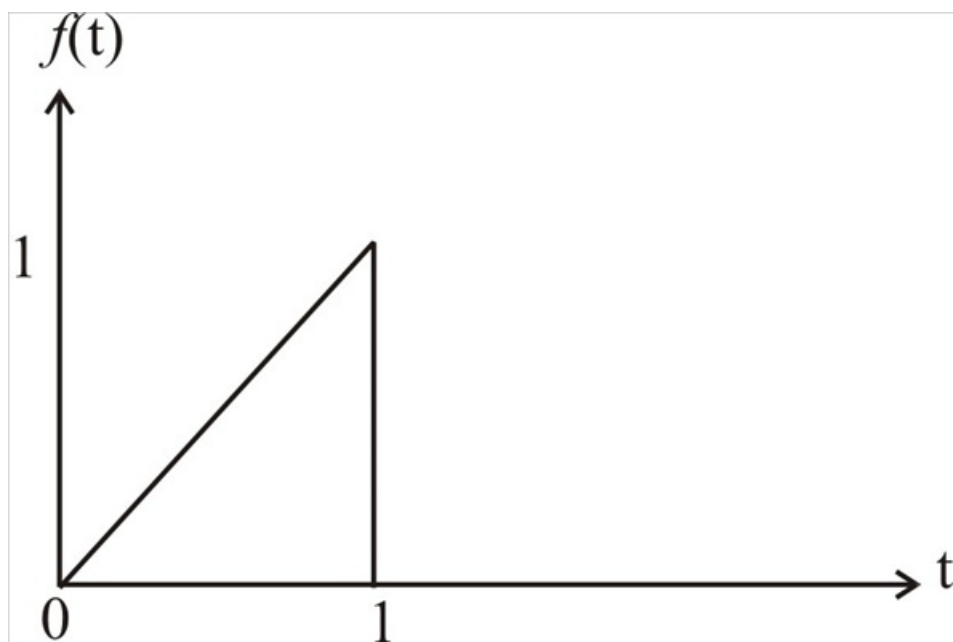
Using Fig. 18.27, design a problem to help other students better understand the Fourier transform given a wave shape.

Figure 18.27



Step-by-step solution

Step 1 of 2



Step 2 of 2

we have

$$f(\omega) = \int_0^1 A e^{-j\omega t} dt$$

$$\Rightarrow f(\omega) = \int_0^1 e^{-j\omega t} dt \quad [A = 1]$$

$$= \frac{1}{-j\omega} [e^{-j\omega t}]_0^1$$

$$= -\frac{1}{j\omega} [e^{-j\omega} - 1]$$

$$\text{or } \boxed{f(\omega) = \frac{1}{j\omega} (e^{-j\omega} - 1)}$$